UNIT 2: ELECTRICITY, MAGNETISM & SEMICONDUCTORS

Charge, electric fields, potential and field lines

Electric charge

Charge is a basic property of particles. Protons are positive, electrons negative, and neutrons neutral. The SI unit is the coulomb (C).

Elementary charge: $e = 1.6 \times 10^{-19}$ C • Quantization: $Q = n\,e$

- Additivity: total charge is the algebraic sum of the charges; include their signs.
- Conservation: charge can transfer between bodies, but the total charge of an isolated system stays constant.
- Interaction: like charges repel; unlike charges attract.

Coulomb's inverse-square law

For two stationary point charges q_1 and q_2 separated by distance r, force magnitude rises with the product of the charges and falls with r^2.

$$F = \frac{1}{4\pi\varepsilon} \times \frac{|q_1 q_2|}{r^2}$$
$$\text{In vacuum: } F = \frac{1}{4\pi\varepsilon_0} \times \frac{|q_1 q_2|}{r^2} \approx 9 \times 10^9 \frac{|q_1 q_2|}{r^2}$$

ε is the permittivity of the medium; ε_0 is the permittivity of free space.

Electric field & intensity

An electric field is the region where a charge experiences electric force. Field intensity is force per unit positive test charge: $E = F/q$. For a point charge Q: $E = \frac{1}{4\pi\varepsilon_0} Q/r^2$.

E is a vector. SI units: N/C or V/m. A uniform field has the same size and direction everywhere (for example, between parallel plates); a point charge produces a non-uniform field.

Potential and potential difference

Electric potential is work per unit positive charge from a reference point: $V = W/q$. For a point charge, $V = \frac{1}{4\pi\varepsilon_0} Q/r$. Potential difference is work per unit charge moved between two points.

1 volt = 1 joule / 1 coulomb • Conventional current flows from higher to lower potential.

ELECTRIC FIELD LINES & MAGNETISM

Use field lines to picture field direction and strength

Electric lines of force

An electric field line is an imaginary curve whose tangent gives the field direction at each point.

- Lines start on positive charges and end on negative charges.
- They never cross or intersect.
- They meet a conductor surface at right angles; in electrostatic equilibrium, the field inside a conductor is zero.
- They can pass through insulators.
- Crowded lines mean a stronger field.
- Uniform-field lines are parallel and equally spaced.

Magnetism

Magnetism is the attraction or repulsion associated with magnets and magnetic materials. A freely suspended bar magnet points roughly north–south. Like poles repel; unlike poles attract. Isolated magnetic poles (monopoles) are not observed.

- Natural magnetite is a natural magnet; magnets can also be made artificially. Uses include motors, bells, speakers, fans, lifting iron and refrigeration.

Magnetic field & intensity

A magnetic field is the region where magnetic force acts. Magnetic field intensity H describes field strength; the notes give SI unit A/m. H is a vector.

Magnetic field lines

- Lines form continuous closed loops and do not intersect.
- Outside a magnet they go from north to south; inside, south to north.
- Closer lines indicate a stronger field. The tangent gives direction at a point.

Magnetic flux: $\Phi = B\,A\cos\theta$ • SI unit: weber (Wb)
Flux density: $B = \Phi/A$ (for a perpendicular field) • SI unit: tesla (T)

$B = \mu H$, where μ is permeability. 1 tesla = 10□ gauss.

CURRENT, OHM'S LAW & MATERIAL PROPERTIES

Core circuit quantities and their units

Electric current

Current is the rate of flow of charge: $I = Q/t$. SI unit: ampere (A). One ampere means one coulomb passes a point each second: 1 A = 1 C/s. Conventional current follows positive-charge flow, from + to −.

Ohm's law

At constant physical conditions, current through a conductor is directly proportional to the potential difference across it.

$$V = I R \bullet R = V/I \bullet 1\ \Omega = 1\ V/A$$

Resistance opposes current; SI unit is ohm (Ω). For an ohmic conductor, the I–V graph is a straight line through the origin.

Conductance, resistivity & conductivity

Quantity	Meaning / formula	SI unit
Conductance G	Reciprocal of resistance: $G = 1/R$	siemens (S)
Resistivity ρ	$\rho = R A / \ell$	ohm metre (Ω·m)
Conductivity σ	Reciprocal of resistivity: $\sigma = 1/\rho$	S/m

A longer conductor has more resistance; a larger cross-sectional area reduces resistance: $R = \rho\ell/A$.

Joule heating

Current through resistance converts electrical energy into heat. The heating effect is used in irons, heaters and geysers.

$$\text{Joule's law: } H = I^2 R t \bullet \text{Also } H = V I t = (V^2/R)t$$

RESISTOR NETWORKS & GALVANOMETERS

Series, parallel and measurement devices

Resistors in series

A single path carries the same current through every resistor. Supply voltage divides among them.

$R_{eq} = R_1 + R_2 + \dots + R_n$

Resistors in parallel

Each branch connects across the same two points, so voltage is the same across each branch and total current splits among branches.

$1/R_{eq} = 1/R_1 + 1/R_2 + \dots + 1/R_n$

Galvanometer

A galvanometer detects small currents. In a moving-coil galvanometer, a current-carrying coil in a magnetic field experiences torque and deflects.

Convert to	How to connect	Added resistance
Ammeter	Low shunt in parallel	$R_{sh} = I_g R_g / (I - I_g)$
Voltmeter	High resistance in series	$R = V/I_g - R_g$

I_g = galvanometer full-scale current; R_g = galvanometer resistance; I = desired ammeter range; V = desired voltmeter range.

Ammeter: very low resistance, connected in series. Voltmeter: very high resistance, connected in parallel across the component.

CONDUCTORS & ENERGY BANDS

Why solids conduct differently

In a solid, valence-band and conduction-band energy levels form bands. The gap between them is the forbidden energy gap Eg; electrons need enough energy to cross it.

Material	Band picture	Typical behaviour
Conductor	Bands overlap or no significant gap	Many mobile electrons; low resistance
Semiconductor	Small forbidden gap	Conductivity between conductor and insulator; rises with temperature
Insulator	Large forbidden gap	Very few mobile carriers; high resistance

Conductors

- Allow charge to flow easily; examples: copper, aluminium, gold and silver.
- High conductivity and low resistivity. Resistance generally rises with temperature for metals.

Semiconductors

- Conductivity lies between conductors and insulators. Examples: silicon (Si) and germanium (Ge).
- A small energy gap lets heat or light free some electrons; conductivity generally increases with temperature.

Insulators

- Strongly resist charge flow; examples: wood, rubber and plastic.
- They have very low conductivity and a large gap between valence and conduction bands.

Band names: valence band = normally bound electrons; conduction band = mobile electrons; forbidden gap = energy range with no allowed states.

INTRINSIC & EXTRINSIC SEMICONDUCTORS

Pure crystals and doping

Intrinsic (pure) semiconductor

An undoped semiconductor such as pure silicon or germanium. Each atom forms covalent bonds with neighbours. At very low temperature it behaves nearly as an insulator; as temperature rises, more electron–hole pairs form.

Extrinsic (doped) semiconductor

Adding a small, controlled impurity to a pure semiconductor increases conductivity. The host is Si or Ge; the added impurity is the dopant.

Type	Dopant	Majority carrier	Examples
P-type	Trivalent acceptor	Holes	B, Al, In
N-type	Pentavalent donor	Electrons	P, As, Sb

P-type

A trivalent impurity has three valence electrons, leaving a bond short of one electron. This vacancy behaves as a hole. Holes are majority carriers; electrons are minority carriers.

N-type

A pentavalent impurity has five valence electrons. Four take part in bonds; the fifth is relatively free. Electrons are majority carriers; holes are minority carriers.

Acceptor doping → P-type → holes dominate.
Donor doping → N-type → electrons dominate.

P–N JUNCTION DIODES

Biasing, I–V behaviour and uses

Junction and depletion region

Joining P-type and N-type material creates a P–N junction. Electrons and holes diffuse across the boundary and recombine, leaving a region depleted of mobile carriers. This creates a potential barrier.

Forward and reverse bias

Bias	Connection	Junction behaviour
Forward	P to +, N to −	Depletion region narrows; current rises after the cut-in voltage
Reverse	P to −, N to +	Depletion region widens; tiny leakage current until breakdown

Approximate cut-in voltage: silicon ≈ 0.7 V; germanium ≈ 0.3 V.

Diode resistance & reverse breakdown

Static resistance: Rdc = Vf/If. Dynamic resistance: Rac = ΔVf/ΔIf. In reverse bias, a large enough voltage can cause breakdown and a sharp current rise.

Applications

- Switching in logic circuits; clipping and clamping circuits; voltage multipliers and regulators.
- Rectification: converts AC to DC. A half-wave rectifier passes one half-cycle and blocks the other.